EE2001 Electrical Circuits- I:

New Topic:

Response of First-Order RL and RC Circuits

4 December 2023

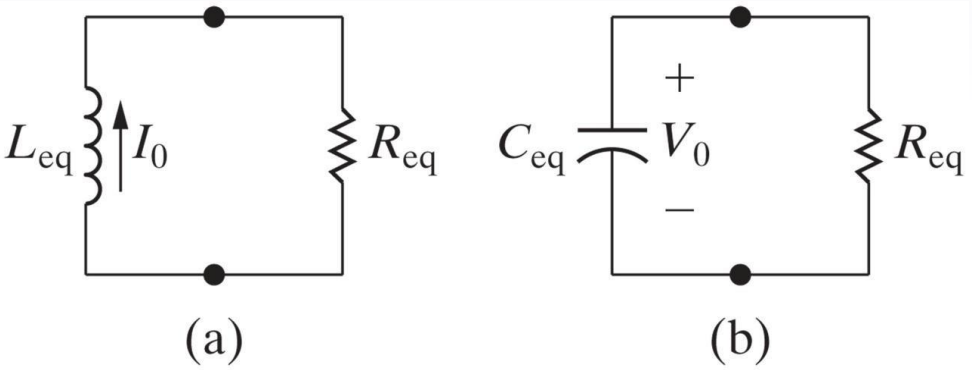
Outline



1. **Introduction**
2. **Natural Response of an RL Circuit**
3. **Natural Response of an RC Circuit**

Introduction

- Circuits which contain (or can be reduced to contain) only one inductor or one capacitor and one resistor are called RL or RC circuits.

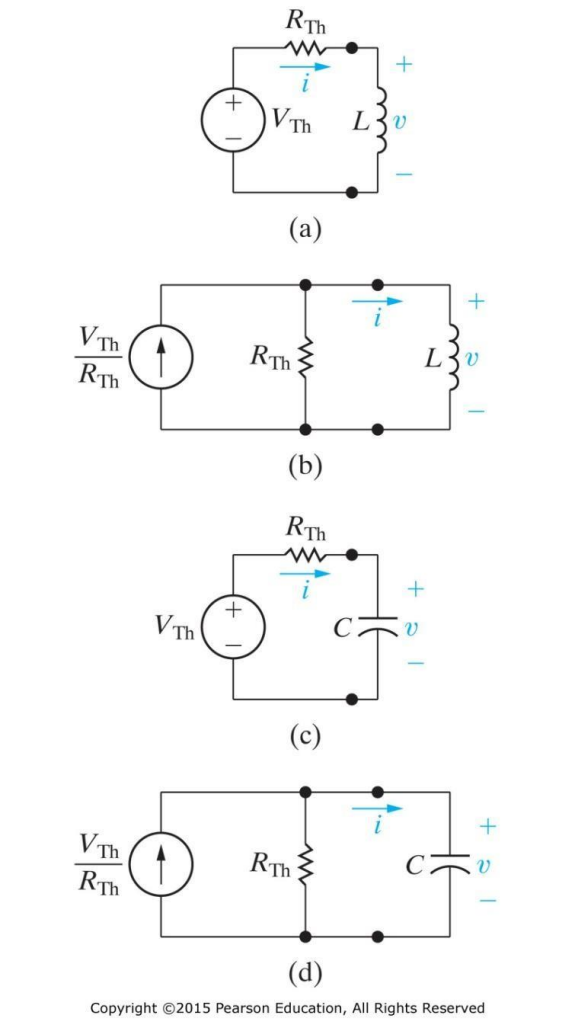


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- Such circuits are also called first order circuits since their response can be described by first order differential equations.

Introduction

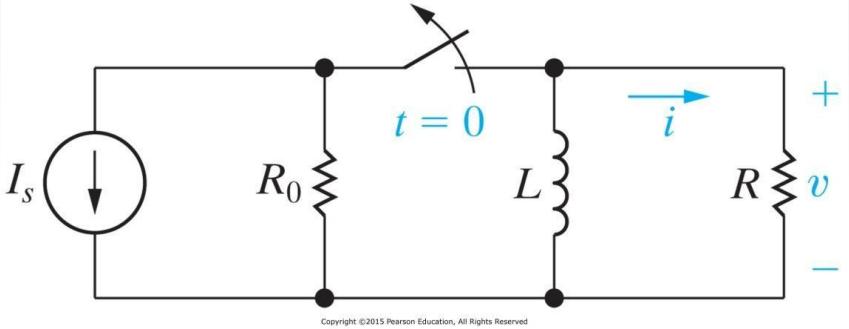
- It is possible to have first order circuits in various combinations
  - Parallel RL or RC circuits
  - Series RL or RC circuits



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The Natural Response of an RL Circuit

- The natural response of a system/circuit is the response to its initial conditions (i.e., no external sources of excitation in the system).
- Consider the circuit



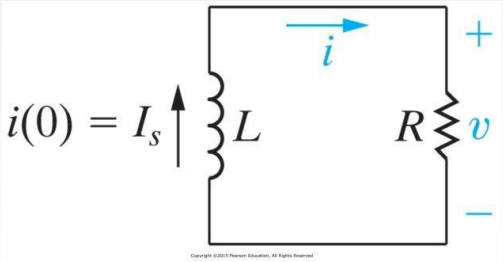
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- Assume that the switch has been closed for a long time (i.e., the currents and voltages have reached constant values) and instantly opened at time $t = 0$.
- Under constant currents, the inductor behaves as a short circuit. This can be used to calculate the initial currents and voltages in the circuit.

  - The initial current through the inductor is $i(0) = I_0 = I_s$.

The Natural Response of an RL Circuit

- After the switch is opened the circuit becomes



- We can write a loop (KVL) equation to solve for the response.

$$L\frac{di}{dt} + Ri = 0$$

- Solving this differential equation one obtains

$$i(t) = i(0)e^{-\frac{R}{L}t} = I_0 e^{-\frac{R}{L}t} = I_s e^{-\frac{R}{L}t}$$

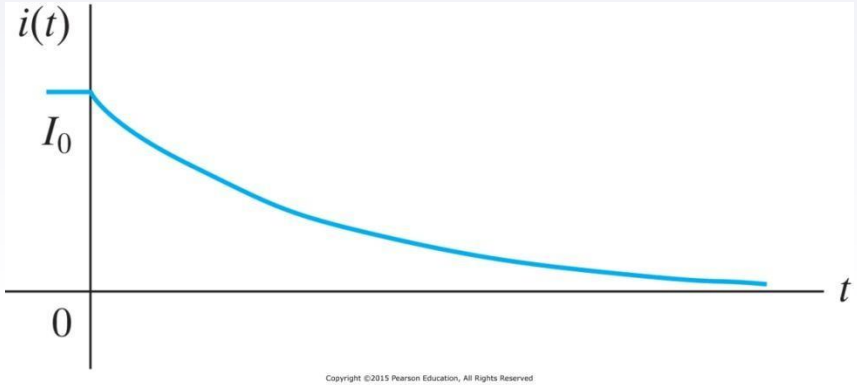
where $i(0) = I_0 = I_s$ is the initial current through the inductor.

- Then, the voltage v can be expressed as

$$v(t) = Ri(t) = RI_0 e^{-\frac{R}{L}t}$$

The Natural Response of an RL Circuit

- The plot of the current becomes as follows



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$$i(t) = i(0)e^{-\frac{R}{L}t} = I_0 e^{-\frac{R}{L}t}$$

- Note that the current in the inductor is a continuous variable and we have $i(0^-) = i(0^+) = I_0$
- $i(t) = I_0 e^{-\frac{R}{L}t}, \qquad t \geq 0$
- $v(0^-) = 0, \qquad v(0^+) = RI_0$
- $v(t) = Ri(t) = RI_0 e^{-\frac{R}{L}t}$

The Natural Response of an RL Circuit

- The power dissipated by the resistor can be calculated as

$$p(t) = RI_0{}^2 e^{-2\frac{R}{L}t}, \qquad t \geq 0^+$$

- Similarly, the energy delivered to the resistor from time 0 to time t can be calculated as

$$w(t) = \int_{t_o}^{t} v(\tau)d\tau = \cdots = \frac{1}{2}L\, I_0{}^2\left(1 - e^{-2\frac{R}{L}t}\right), \qquad t \geq 0^+$$

- Note that as $t \to \infty$ we have

$$w(t) \to \frac{1}{2}LI_0{}^2$$

- The total energy delivered to the resistor is the total energy initially stored in the inductor.
- The total energy delivered to the resistor is independent on its resistance value.
- The value of the resistor affects only the rate in which the energy is delivered through the term (R/L).

Time Constant

- The value of (R/L) determines also the rate at which the current and the voltage approach their final values (i.e., zero for the circuit considered above).
- The reciprocal of (R/L) is defined as the time constant

$$\tau = \frac{L}{R}$$

- Using the time constant, the response equations become

$$i(t) = I_0 e^{-t/\tau}$$

$$v(t) = R I_0 e^{-t/\tau}$$

$$p(t) = R {I_0}^2 e^{-2t/\tau}$$

$$w(t) = \frac{1}{2} L {I_0}^2 \left(1 - e^{-2t/\tau}\right)$$

- The time constant determines the rate at which a signal approaches its final value.

Time Constant

- The value of $e^{-t/\tau}$ for various values of t are given below.

TABLE 7.1 Value of $e^{-t/\tau}$ For t Equal to Integral Multiples of τ

t	$e^{-t/\tau}$	t	$e^{-t/\tau}$
τ	3.6788×10^{-1}	6τ	2.4788×10^{-3}
2τ	1.3534×10^{-1}	7τ	9.1188×10^{-4}
3τ	4.9787×10^{-2}	8τ	3.3546×10^{-4}
4τ	1.8316×10^{-2}	9τ	1.2341×10^{-4}
5τ	6.7379×10^{-3}	10τ	4.5400×10^{-5}

- In $t = 5\tau$ time the signal is close to more than 99% of its final (steady state) value.
- Therefore, $t = 5\tau$ time is a sufficiently long time for the signal to get close to its final value.
- The response for small t ($t < 5\tau$) can be referred to as transient response.
- The response for large t ($t > 5\tau$) can be referred to as steady state response.

Time Constant

- The time constant is an important characteristic of any first order linear circuit/system.
- If the time constant is not known but the plot of the response is known it can be determined from the plot.
- Consider the equation of the current
- $i(t) = I_0 e^{-t/\tau}$ $\qquad \frac{di}{dt} = -\frac{I_0}{\tau} e^{-t/\tau}$
- At $t = 0^+$ the value of the derivative is

$$\frac{di}{dt}(0^+) = -\frac{I_0}{\tau}$$

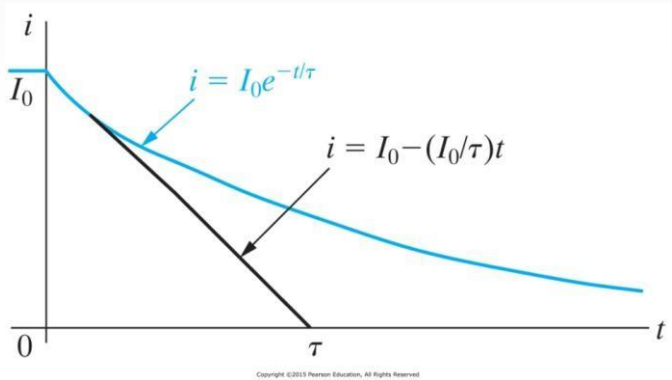
- This is the slope of the tangent to the response curve at $t = 0$.

Time Constant

- The equation of the tangent can be written as

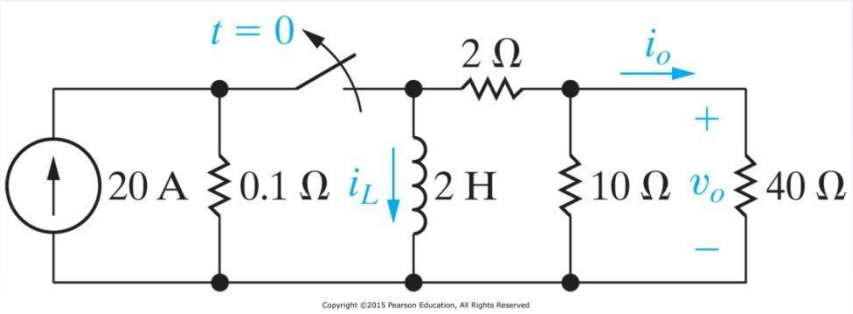
$$i = I_0 - \frac{I_0}{\tau}$$

- The tangent line to the response plot at t = 0 crosses the horizontal axis at $t = \tau$.



Time Constant

- Example-1: The switch in the given circuit has been closed for a long time before it is opened at $t = 0$. Find



a) $i_L(t)$ for $t \geq 0$,

b) $i_0(t)$ for $t \geq 0^+$,

c) $v_0(t)$ for $t \geq 0^+$,

d) the percentage of the total energy stored in the 2 H inductor that is dissipated in the 10 Ω resistor.

Solution: a) Because the switch has been closed for a long time prior to $t = 0$, we know that the voltage across inductor must be zero at $t = 0^-$. Therefore the initial current in the inductor is 20 A at $t = 0^-$. Hence, $i_L(0^+)$ also is 20 A, because instantaneous change in the current cannot occur in an inductor. We replace the resistive circuit connected to the terminals of the inductor with a single resistor of 10 Ω.

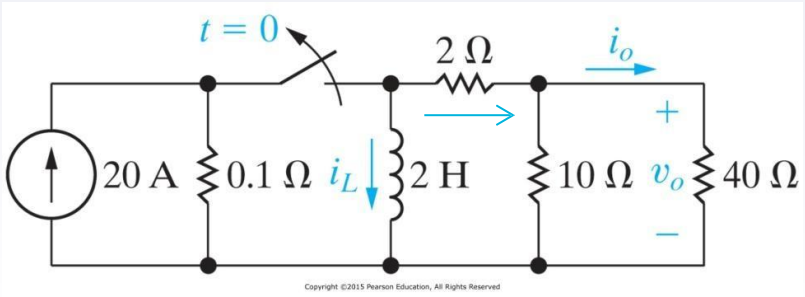
$$R_{eq} = 2 + (40||10) = 10\,\Omega$$

Time Constant

- Example-1: The switch in the given circuit has been closed for a long time before it is opened at $t = 0$. Find

  a) $i_L(t)$ for $t \geq 0$,

  b) $i_0(t)$ for $t \geq 0^+$,



Continuation of a) The time constant of the circuit is L/R_{eq}, or 0.2 s, giving the expression for the inductor current as

$$i_L(t) = 20e^{-5t}\ A, \qquad t \geq 0$$

b) We find the current in the 40 Ω resistor most easily by using current division:

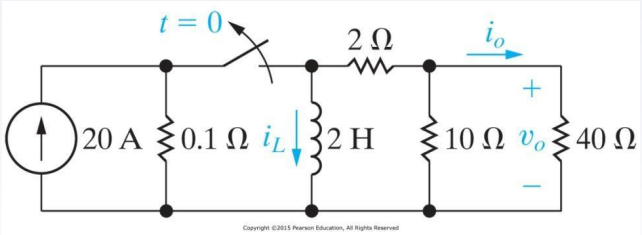
$$i_0 = -i_L \frac{10}{10 + 40}$$

This expression is valid for $t \geq 0^+$ because $i_0 = 0$ at $t = 0^-$. The inductor behaves as a short circuit prior to the switch being opened, producing an instantaneous change in the current i_0. Then,

$$i_0(t) = -4e^{-5t}, \qquad t \geq 0^+$$

Time Constant

- Example-1: The switch in the given circuit has been closed for a long time before it is opened at $t = 0$. Find



  c) $v_0(t)$ for $t \geq 0^+$,

  d) the percentage of the total energy stored in the 2 H inductor that is dissipated in the 10 Ω resistor.

Solution: c) We find the voltage v_0 by direct application of Ohm's law:

$$v_0(t) = 40i_0 = -160e^{-5t}\ V, \qquad t \geq 0^+$$

d) The power dissipated in the 10 Ω resistor is

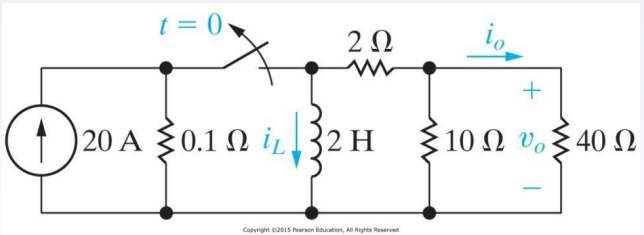
$$p_{10\Omega}(t) = \frac{{v_o}^2}{R} = \frac{{v_o}^2}{10} = 2560e^{-10t}\ W, \qquad t \geq 0^+$$

The total energy dissipated in the 10 Ω resistor is

$$w_{10\Omega}(t) = \int_0^\infty 2560e^{-10t}dt = 256\ J$$

Time Constant

- Example-1: The switch in the given circuit has been closed for a long time before it is opened at $t = 0$. Find



  c) $v_0(t)$ for $t \geq 0^+$,

  d) the percentage of the total energy stored in the 2 H inductor that is dissipated in the 10 Ω resistor.

Continuation of d) The total energy dissipated in the 10 Ω resistor is

$$w_{10\Omega}(t) = \int_0^\infty 2560e^{-10t}dt = 256\,J$$

The initial energy stored in the 2 H inductor is

$$w(0) = \frac{1}{2}Li^2(0) = \frac{1}{2}(2)(400) = 400\,J$$

Therefore the percentage of energy dissipated in the 10 Ω resistor is

$$\frac{256}{400}(100) = 64\%$$

Time Constant

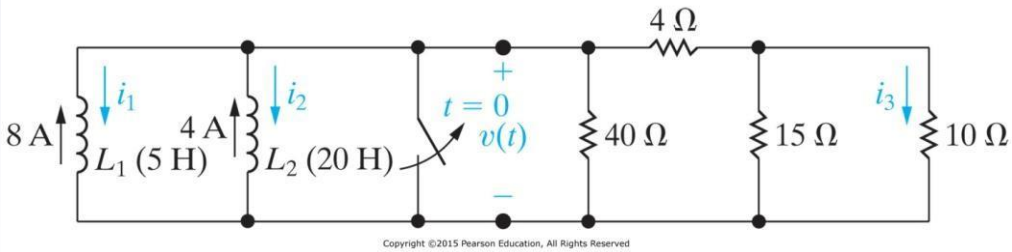
- Example-2: In the given circuit, the initial current in inductors L_1 and L_2 have been established by sources not shown. The switch is opened at $t = 0$.

a) Find i_1, i_2 and i_3 for $t \geq 0$,

b) Calculate the initial energy stored in the parallel inductors.

c)Determine how much energy is stored in the inductors as $t \to \infty$

d)Show that the total energy delivered to the resistive network equals the difference between the results obtained in (b) and (c).

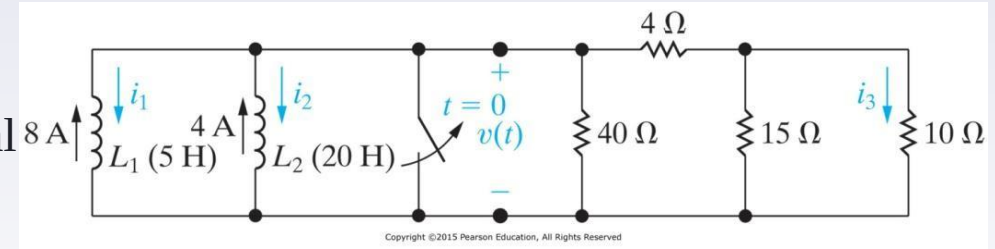


a) The key to finding currents i_1, i_2 and i_3 lies in knowing the voltage $v(t)$. We can easily find $v(t)$ by reducing the circuit topology.

The parallel inductors simplify to an equivalent inductance of 4 H, carrying an initial current of 12 A. The resistive network reduces to a single resistance of 8 Ω.

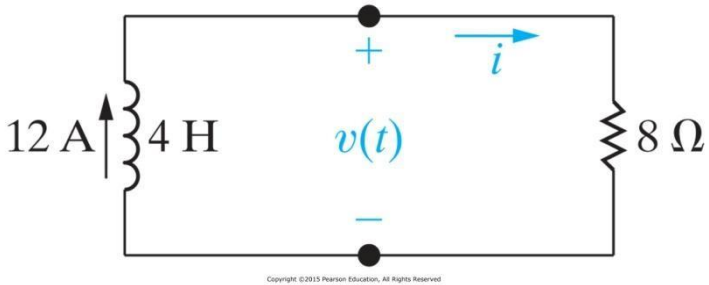
Time Constant

- Example-2: In the given circuit, the initial current in inductors L_1 and L_2 have been established by sources not shown. The switch is opened at $t = 0$.



a) Find i_1, i_2 and i_3 for $t \geq 0$,

b) Calculate the initial energy stored in the parallel inductors.



Continuation of a)

The initial value of $i(t)$ is 12 A and the time constant is 4/8, or 0.5 s. Therefore,

$$i(t) = 12e^{-2t}\ A, \qquad t \geq 0$$

Now $v(t)$ is simply the product of $8i$, so

$$v(t) = 96e^{-2t}\ A, \qquad t \geq 0^+$$

The circuit shows that $v(t) = 0$ at $t = 0^-$, so the expression for $v(t)$ is valid for $t \geq 0^+$.

After obtaining $v(t)$, we can calculate i_1, i_2 and i_3.

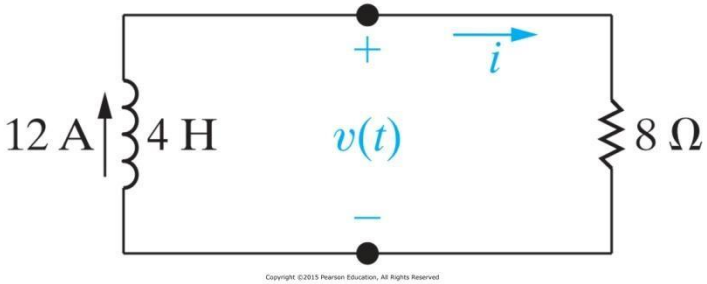
$$i_1 = \frac{1}{5}\int_0^t 96e^{-2x}dx - 8 = 1.6 - 9.6e^{-2t}\ A, \qquad t \geq 0$$

Time Constant

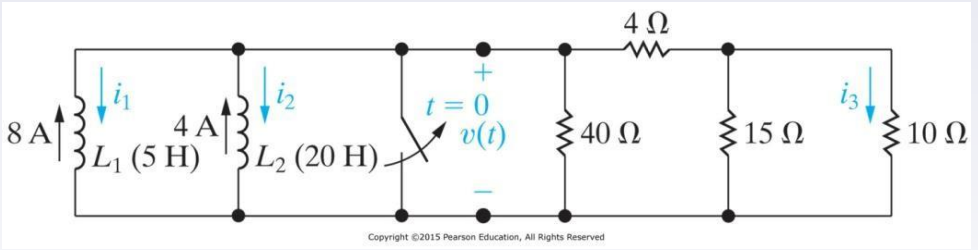
- Example-2: In the given circuit, the initial current in inductors L_1 and L_2 have been established by sources not shown. The switch is opened at $t = 0$.

a) Find i_1, i_2 and i_3 for $t \geq 0$,

b) Calculate the initial energy stored in the parallel inductors.



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Continuation of a)

$$i_2 = \frac{1}{20}\int_0^t 96e^{-2x}dx - 4\ A$$

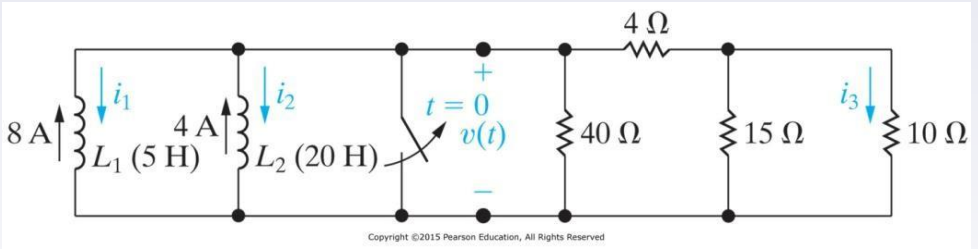
$$= 1.6 - 2.4e^{-2t}\ A, \qquad t \geq 0$$

$$i_3 = \frac{v(t)}{10}\frac{15}{25} = 5.76e^{-2t}\ A, \qquad t \geq 0^+$$

Note that the expressions for the inductor currents i_1 and i_2 are valid for $t \geq 0$, whereas the expression for the resistor current i_3 is valid for $t \geq 0^+$.

Time Constant

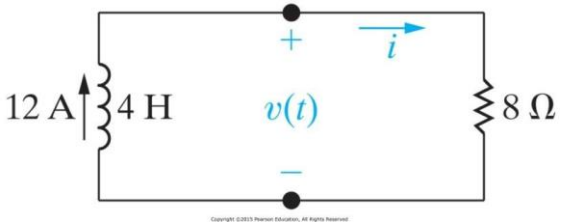
- Example-2: In the given circuit, the initial current in inductors L_1 and L_2 have been established by sources not shown. The switch is opened at $t = 0$.



b) Calculate the initial energy stored in the parallel inductors.

c)Determine how much energy is stored in the inductors as $t \to \infty$

d)Show that the total energy delivered to the resistive network equals the difference between the results obtained in (b) and (c).



b) The initial energy stored in the inductors is

$$w = \frac{1}{2}(5)(64) + \frac{1}{2}(20)(16) = 320\,J$$

c) As $t \to \infty$, $i_1 \to 1.6\,A$ and $i_2 \to 1.6\,A$. Therefore, a long time after the switch has been opened, the energy stored in the two inductors is

$$w = \frac{1}{2}(5)(1.6)^2 + \frac{1}{2}(20)(-1.6)^2 = 32\,J$$

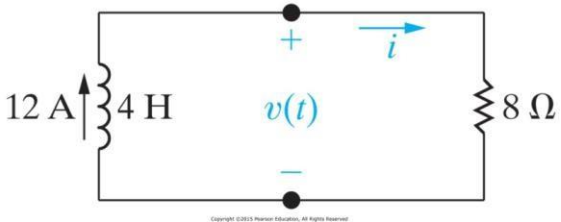
Time Constant

- Example-2: In the given circuit, the initial current in inductors L_1 and L_2 have been established by sources not shown. The switch is opened at $t = 0$.

b) Calculate the initial energy stored in the parallel inductors.

c)Determine how much energy is stored in the inductors as $t \to \infty$

d)Show that the total energy delivered to the resistive network equals the different between the results obtained in (c) and (d).



d) We obtain the total energy delivered to the resistive network by integrating the expression for the instantaneous power from zero to infinity:

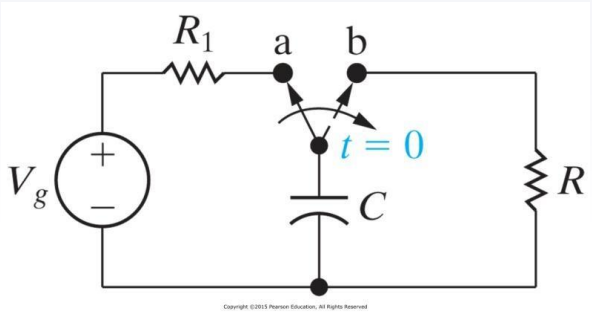
$$w = \int_0^\infty p\,dt = \int_0^\infty 1152e^{-4}dt$$

$$= 1152\frac{e^{-4}}{-4}\Big|_0^\infty = 288\,J$$

This result is the difference between the initially stored energy (320 J) and the energy trapped in the parallel inductors (32 J). The equivalent indıctor for the parallel inductors has initial energy of 288 J; that is the energy stored in the equivalent inductor represents the amount of energy that will be delivered to the resistive network at the terminals of the original inductors.

The Natural Response of an RC Circuit

- The natural response of an RC circuit can be obtained in a similar way to the natural response of an RL circuit.
- Consider the circuit

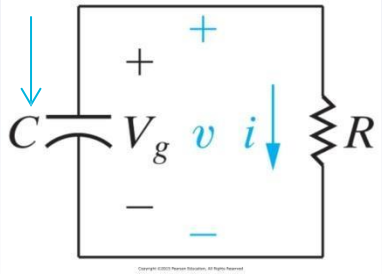


- Assume that the switch has been at location “a” for a long time (i.e., the currents and voltages have reached constant values) and is instantly moved to position “b” at time $t = 0$.
- Under constant currents the capacitor behaves as an open circuit. This can be used to calculate the initial currents and voltages in the circuit.

  - The initial voltage across the capacitor is $v(0) = V_0 = V_g$

The Natural Response of an RC Circuit

- After the switch is opened the circuit becomes



- We can write a node (KCL) equation to solve for the response.

$$C\frac{dv}{dt}+\frac{v}{R}=0$$

- Solving this differential equation one obtains

$$v(t)=v_{(}0_{)}e^{-\frac{t}{RC}}=V_{0}e^{-\frac{t}{RC}}=V_{g}e^{-\frac{t}{RC}}$$

where $v(0)=V_0=V_g$is the initial voltage across the capacitor.

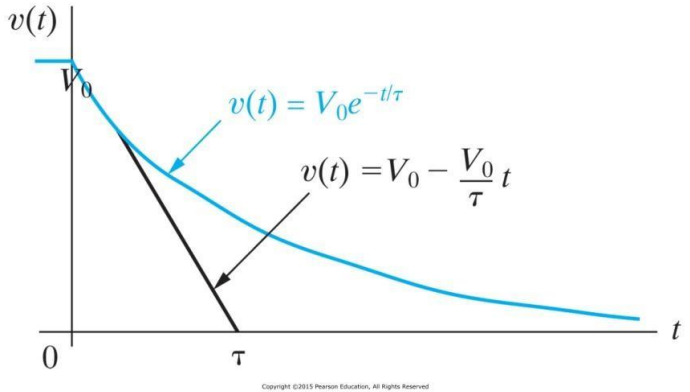
The Natural Response of an RC Circuit

- The time constant for an RC circuit is $\tau = RC$
- The current and voltage responses using the time constant are

$$v(t) = V_0 e^{-t/\tau}$$

$$i(t) = \frac{V_0}{R} e^{-t/\tau}$$

- Similar to the case of an RL circuit the time constant can be determined using
  - the tangent to the plot of the response



- $\frac{dv}{dt}(0^+) = -\frac{V_0}{\tau}$
- $v = V_0 - \frac{V_0}{\tau} t$
- dt

The Natural Response of an RC Circuit

- The power dissipated by the resistor can be calculated as

$$p(t) = vi = \frac{{V_0}^2}{R} e^{-2t/\tau}$$

- Similarly, the energy delivered to the resistor from time 0 to time t can be calculated as

$$w(t) = \int_0^t p(\tau)\, d\tau = \cdots = \frac{1}{2} C {V_0}^2 \left(1 - e^{-2t/\tau}\right)$$

- Note that as $t \to \infty$ we have

$$w(t) \to \frac{1}{2} C {V_0}^2$$

- The energy delivered to the resistor is the total energy initially stored in the capacitor.
- The energy delivered to the resistor is independent on its resistance value.
- The value of the resistor affects only the rate in which the energy is delivered through the time constant $\tau = RC$

The Natural Response of an RC Circuit

3

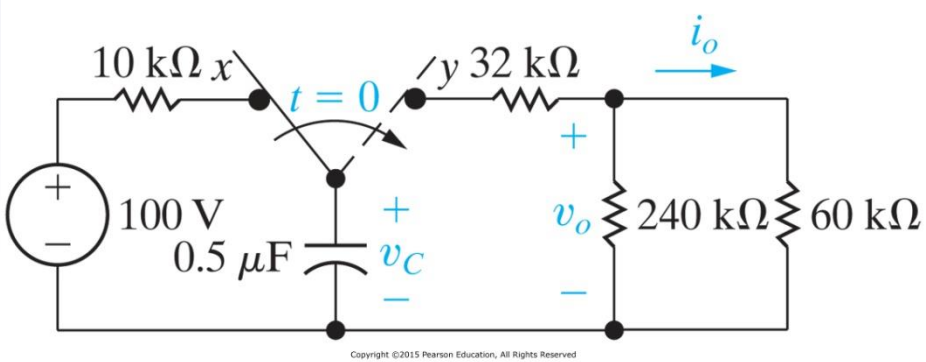
- Example-3: The switch in the circuit has been in position x for a long time. At $t = 0$, the switch moves instantaneously to position y. Find

  a) $v_C(t)$ for $t \geq 0$,

  b) $v_0(t)$ for $t \geq 0^+$,

  c) $i_0(t)$ for $t \geq 0^+$,

  d) the total energy dissipated in the 60 kΩ resistor.



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Solution a) Because the switch has been in position x for a long time, the 0.5 mF capacitor will charge to 100 V and be positive at the upper terminal. We can replace the resistive network (32+240||60) kΩ connected to the capactior at $t = 0^+$ with an equivalent resistance of 80 kΩ. Hence the time constant of the circuit is $(0.5 \times 10^{-6})(80 \times 10^3)$ or 40 ms. Then,

$$v_c(t) = V_o e^{-t/(RC)} = 100e^{-25t}\ V, \qquad t \geq 0$$

The Natural Response of an RC Circuit

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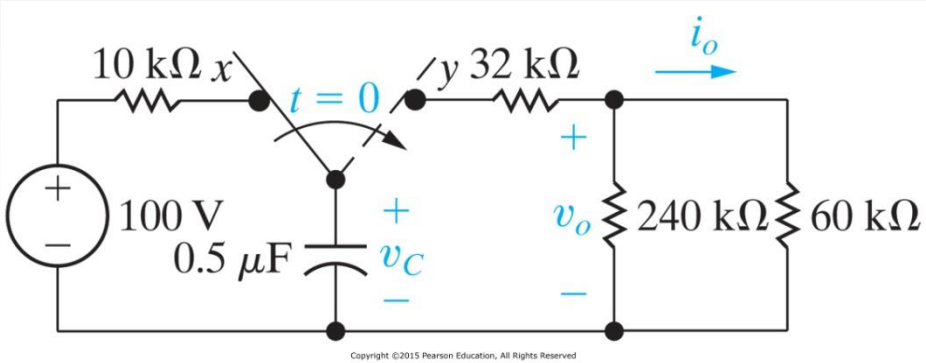
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  a) $v_C(t)$ for $t \geq 0$,

  b) $v_0(t)$ for $t \geq 0^+$,

  c) $i_0(t)$ for $t \geq 0^+$,

  d) the total energy dissipated in the 60 kΩ resistor.



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Solution b) The easiest way to find $v_0(t)$ is to note that the resistive circuit form a voltage divider across the terminals of the capacitor. Thus,

$$v_0(t) = \frac{48}{80} v_C(t) = 60e^{-25t}\ V, \qquad t \geq 0^+$$

This expression for $v_0(t)$ is valid for $t \geq 0^+$ because $v_0(0^-)$ is zero. Thus we have an instantaneous change in the voltage across the 240 k Ω resistor.

The Natural Response of an RC Circuit

3

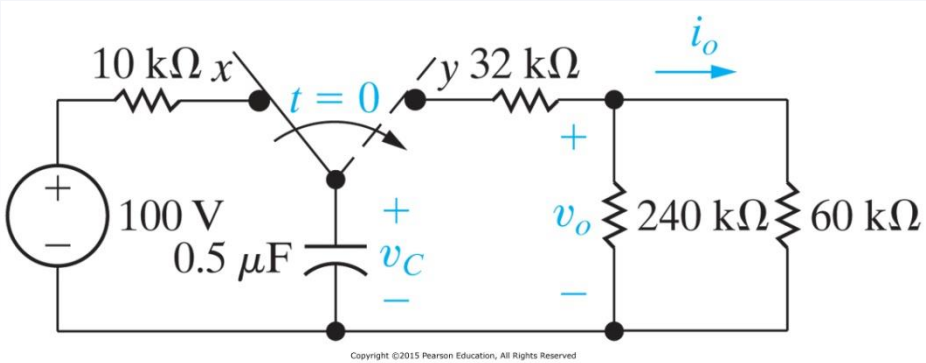
- Example-3: The switch in the circuit has been in position x for a long time. At $t = 0$, the switch moves instantaneously to position y. Find

  a) $v_C(t)$ for $t \geq 0$,

  b) $v_0(t)$ for $t \geq 0^+$,

  c) $i_0(t)$ for $t \geq 0^+$,

  d) the total energy dissipated in the 60 kΩ resistor.



Solution c) We find the current $i_0(t)$ from Ohm's law:

$$i_0(t) = \frac{v_0(t)}{6010^3} = e^{-25t}\ mA = e^{-25t} \times 10^{-3}\ A \qquad t \geq 0^+$$

d) The power dissipated in the 60 kΩ resistor: $p_{60k\Omega}(t) = {i_0}^2(t)(60 \times 10^3) = 60e^{-50t}\ mW$

The total energy dissipated is: $w_{60k\Omega} = \int_0^\infty {i_0}^2(t)(60 \times 10^3)dt = 1.2\ mJ$

End of the Lesson (04.12.23)

Any questions?

Supplementary videos: Enter Youtube-> Search for ‘Alparslan Zehir’ -> Go to Playlists -> Watch ‘Electrical Circuits’ playlist